

Assignment 2: Motion Control

Autonomous Driving Systems - Planning and Control Module

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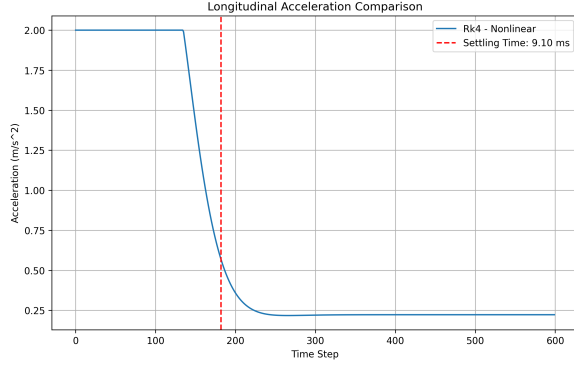
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Introduction

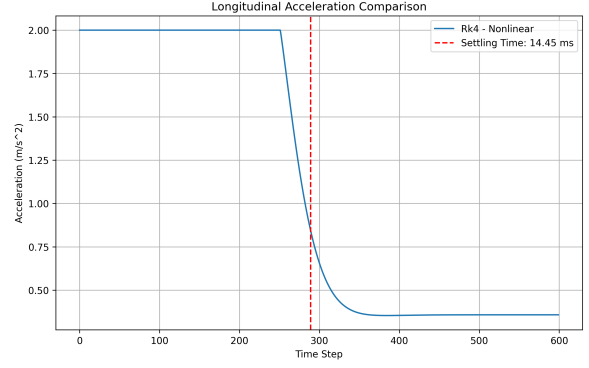
This report presents the tests performed on the tuned controllers, along with the correspondent achieved results. Different types of controller have been used in this assignment: PID was used for the longitudinal control; for the lateral control, three different approaches have been used instead: Pure Pursuit, Stanley and MPC.

1 Exercise 1: Longitudinal Control

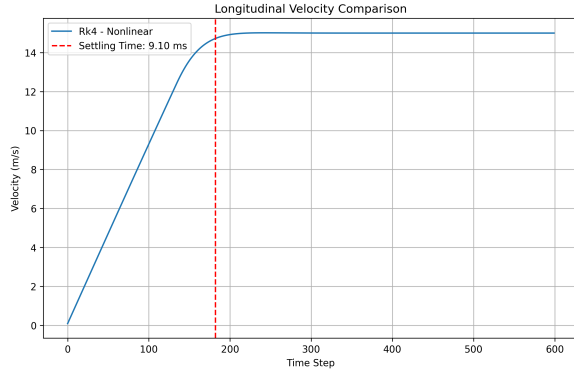
The purpose of this exercise was to tune the PID controller in order to make the vehicle reach a certain velocity (i.e. 15 m/s and 25 m/s). After some tests, the following values for the PID have been chosen: $k_p = 1.6$; $k_i = 0.85$; $k_d = 0.01$.



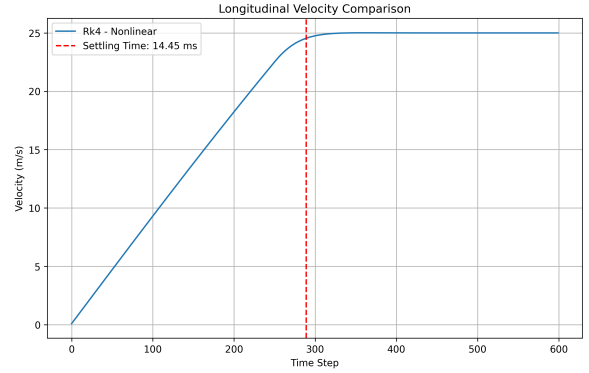
(a) a_x for target $v = 15$ m/s



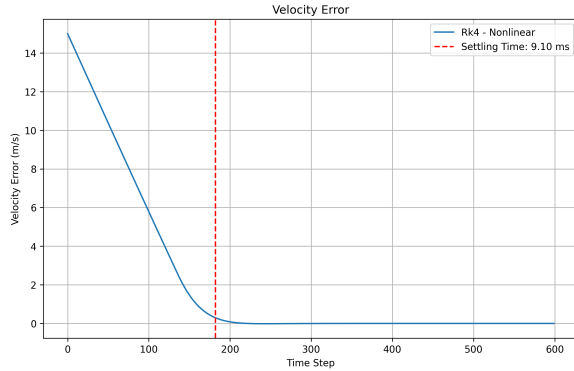
(b) a_x for target $v = 25$ m/s



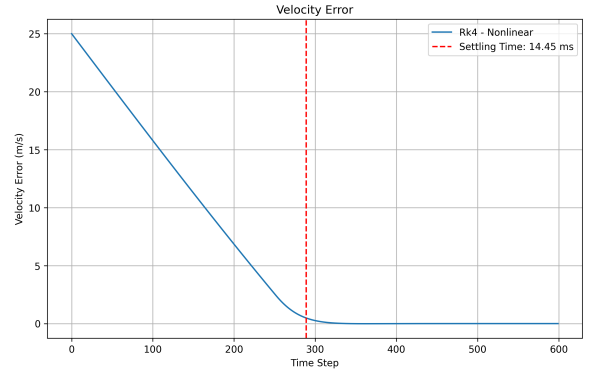
(c) v_x for target $v = 15$ m/s



(d) v_x for target $v = 25$ m/s



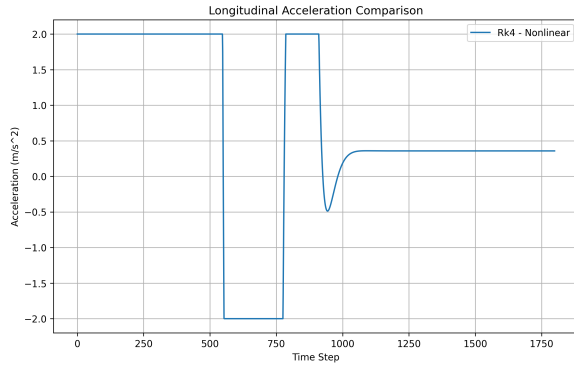
(e) $e(t)$ for target $v = 15$ m/s



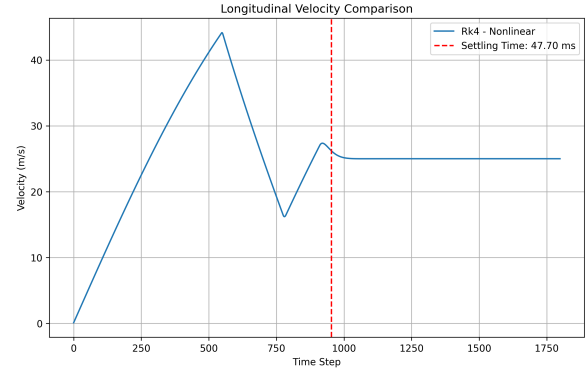
(f) $e(t)$ for target $v = 25$ m/s

Figure 1: PID responses at different target velocities

As shown in Figure 1, the controller response is critically damped, allowing the vehicle to reach the target speed with a low settling time in both cases. The behavior of the controller is similar in both cases: an initial period of maximum acceleration (i.e. 2 m/s) is followed by a steep decrease of it until a certain value greater than zero necessary to contrast the contrasting forces (Fig. 1a, 1b). Despite the omission of the derivative term k_p , the responses are characterized by zero overshoot and steady state error: this is due to the implemented anti windup strategies, which prevent the accumulation of the error by the integral term. Indeed, Figure 2 shows the response in a test performed without the anti windup system with the same PID configuration: a clear overshoot problem can be noticed, despite the system being able to stabilize itself after some oscillations.



(a) a_x for target $v = 25$ m/s



(b) v_x for target $v = 25$ m/s

Figure 2: PID response for target speed 25 m/s without the anti windup protection

2 Exercise 2: Low-speed Lateral Control

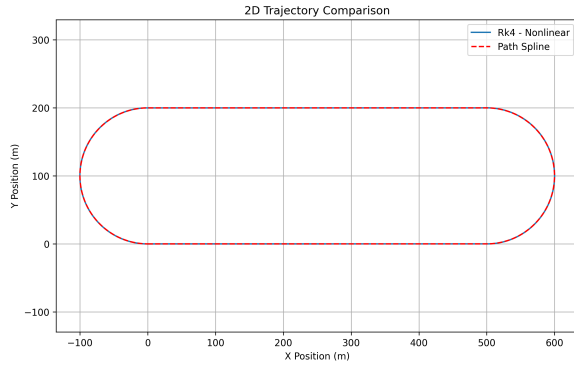
In this exercise, lateral control strategies were introduced and evaluated. The primary objective was to compare the performance of the Stanley and Pure Pursuit controllers at low to medium speeds (10 m/s and 20 m/s). The experimental results demonstrate accurate path tracking for both algorithms, as illustrated in the trajectory plots in Figure 3. Furthermore, both controllers successfully satisfied the design constraints for lateral and velocity errors (see Figures 4 and 8). The specific tuning parameters adopted for each controller are detailed below:

Pure Pursuit Parameters:

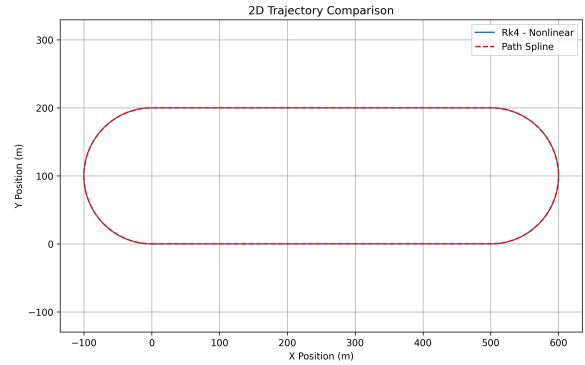
$$k_{pp} = 0.1 \quad (\text{Speed Gain}) \quad L_{d,min} = 2.0 \text{ m} \quad (\text{Min Look-ahead})$$

Stanley Parameters:

$$k_{stanley} = 1.9 \quad (\text{CTE Gain}) \quad W_{cte} = 9.5 \quad (\text{CTE Weight}) \quad W_{yaw} = 1.9 \quad (\text{Heading Weight})$$

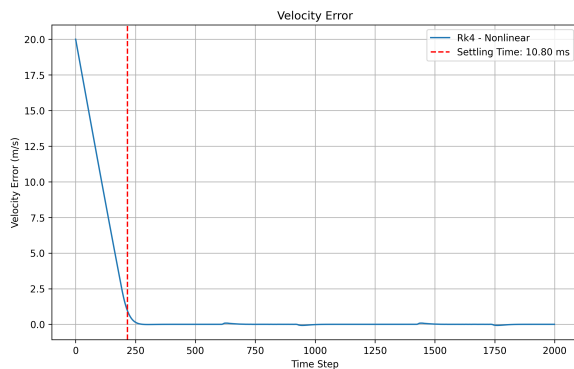


(a) Pure Pursuit

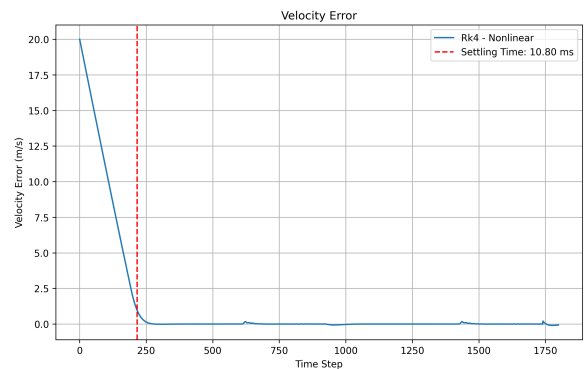


(b) Stanley

Figure 3: Trajectories performed by both controllers at 20 m/s



(a) Pure Pursuit



(b) Stanley

Figure 4: Velocity errors at 20 m/s

Longitudinal control is managed by the PID controller tuned in the previous section, which maintains a critically damped response (Fig. 5). However, minor acceleration oscillations are observable after the settling time. These are attributed to the higher operating speeds compared to the previous exercise: the increased lateral dynamics generate

significant drag forces during cornering, which the longitudinal controller must actively compensate for to maintain the target velocity.

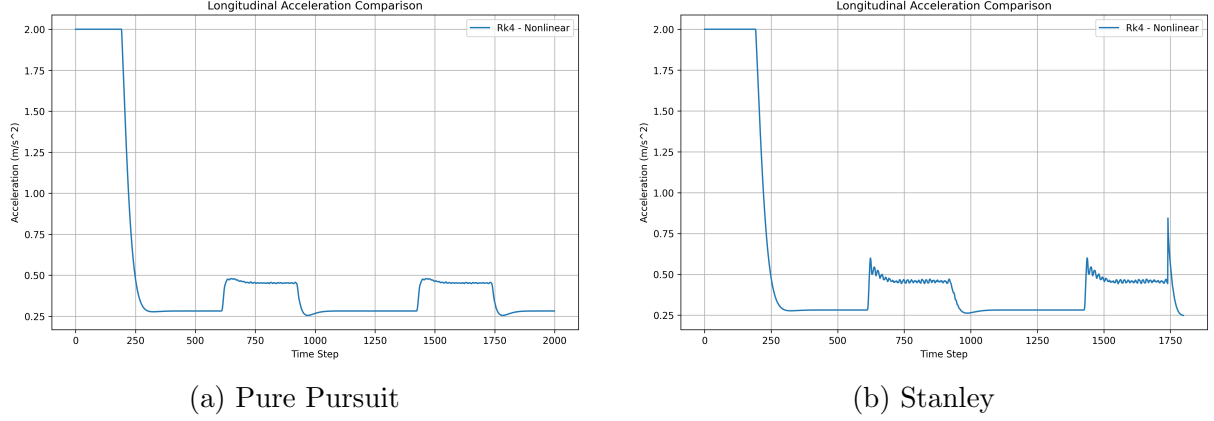


Figure 5: a_x comparison with a target velocity of 20 m/s

2.1 Pure Pursuit corner-cutting behavior

The experimental results highlight a distinct transition in the vehicle's lateral dynamics as velocity increases, illustrating the conflict between the controller's geometric logic and the vehicle's physical inertia. At low speeds (10 m/s), the Pure Pursuit controller exhibits its characteristic "corner-cutting" behavior, resulting in a negative lateral error as the algorithm geometrically anticipates the path curvature (Fig. 6a). This typical behavior of the Pure Pursuit controller is due to its nature of making a move based on a forward point on the path, rather than the point at the same longitudinal Frenet coordinates, considered instead by Stanley. By this assumption, Pure Pursuit suffers from corner cutting, especially if the look ahead distance is too high. However, at higher velocities (20 m/s and above), this geometric anticipation is overpowered by the vehicle's dynamics. The significant increase in centrifugal force generates substantial tire slip angles (Fig 7), causing the vehicle to understeer and drift towards the outside of the curve. Consequently, the lateral error shifts to positive values (Fig. 8a), demonstrating that at high speeds, the kinematic assumptions of the Pure Pursuit controller are insufficient to counteract the physical tendency of the vehicle to slide outward.

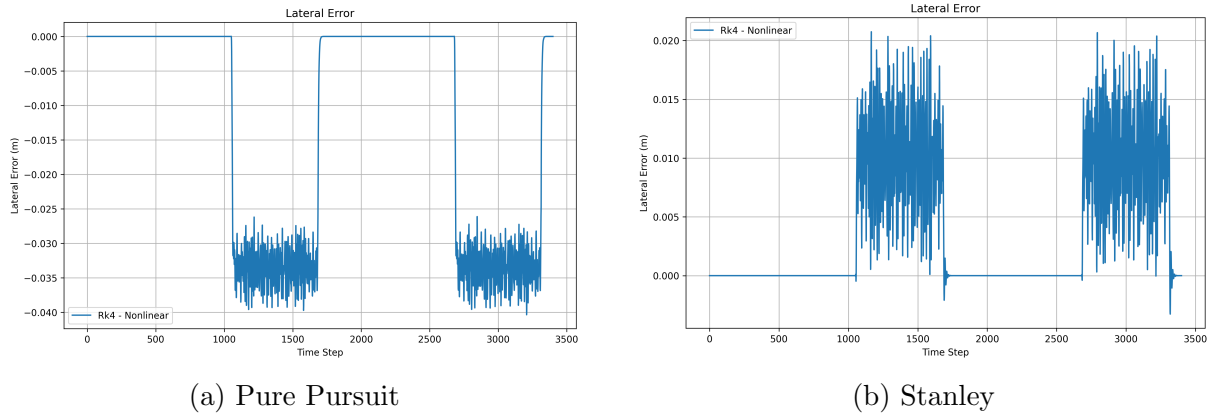
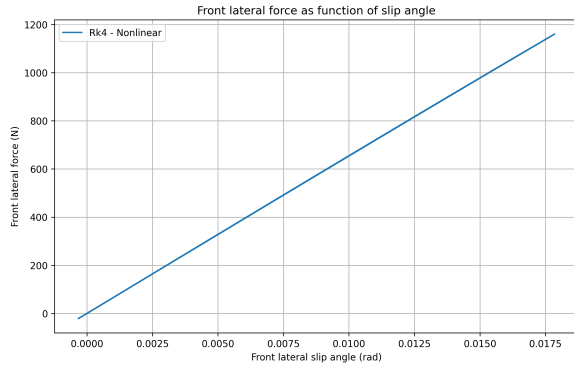
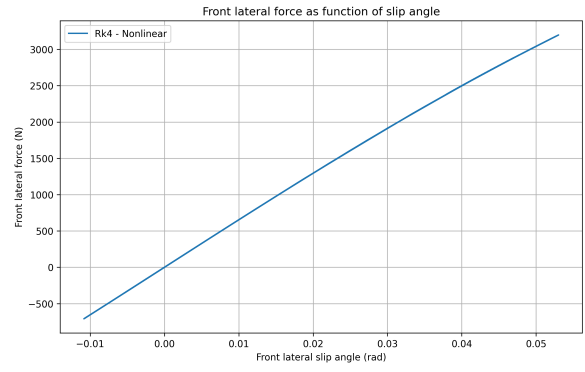


Figure 6: Lateral error over time at 10 m/s

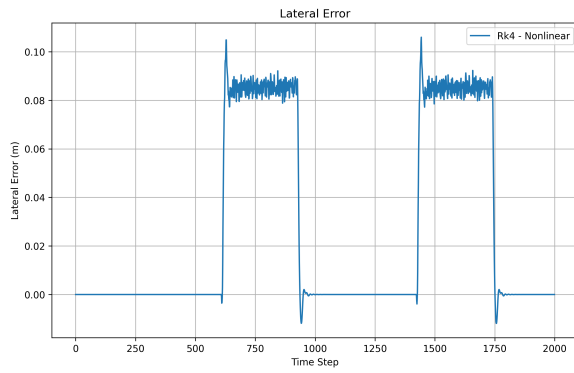


(a) Target velocity 10 m/s

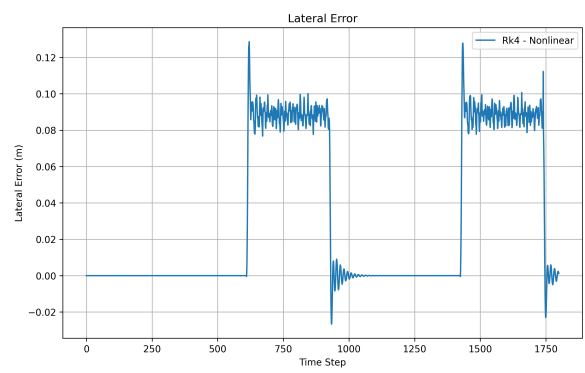


(b) Target velocity 20 m/s

Figure 7: Front Lateral Force vs Front Slip Angle comparison for Pure Pursuit



(a) Pure Pursuit



(b) Stanley

Figure 8: Lateral error over time at 20 m/s

3 Exercise 3: High-speed Lateral Control

The purpose of this exercise was testing both controllers at higher velocities (23 m/s, 25 m/s and at maximum achievable speed respecting both speed error and lateral error constraints). Both Pure Pursuit and Stanley are tuned in the same way presented in the previous section. In addition, MPC controller has been tuned and tested.

3.1 Convergence differences between Pure Pursuit and Stanley

A comparative analysis of the Stanley and Pure Pursuit algorithms reveals a fundamental trade-off between tracking accuracy and control smoothness. The Stanley controller, which computes the steering command based on the instantaneous cross-track error at the front axle, achieves superior convergence to the reference path, maintaining a lower average lateral error throughout the simulation. However, this high reactivity comes at the cost of stability; the lateral error plot for the Stanley controller displays noticeable high-frequency oscillations as shown in Fig 9. In contrast by targeting a point further down the path, the Pure Pursuit algorithm effectively smooths out high-frequency disturbances, resulting in a more stable steering command but a slower response to curvature changes (Fig. 10).

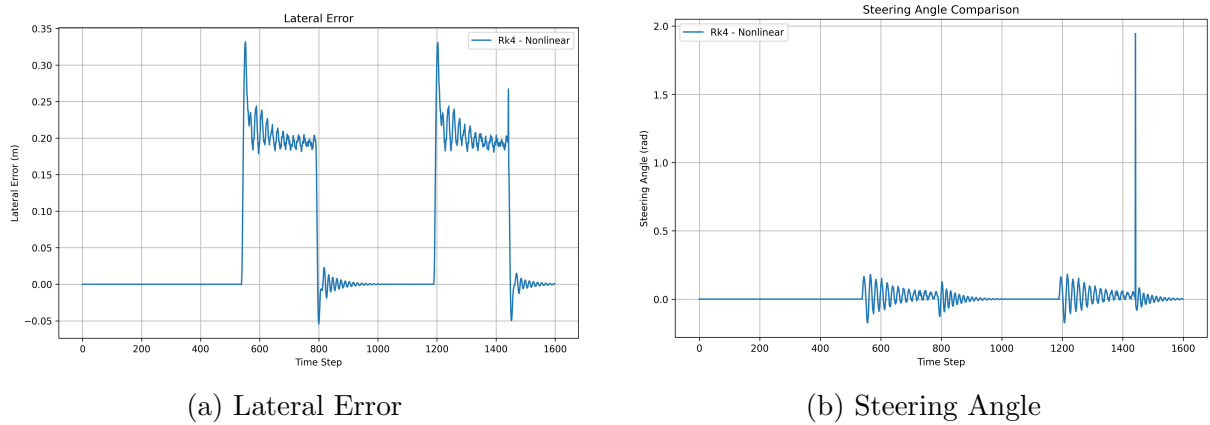


Figure 9: Stanley controller behavior at 25 m/s

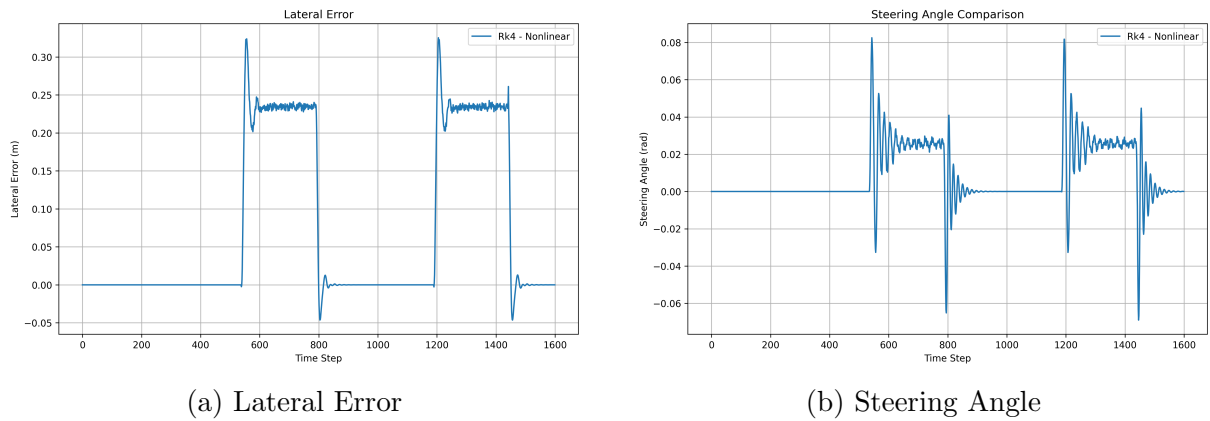


Figure 10: Pure Pursuit controller behavior at 25 m/s

3.2 MPC implementation and achieved results

This stage of the analysis focused on the implementation and testing of a Model Predictive Control (MPC) strategy. MPC represents a superior approach compared to kinematic controllers (such as Stanley and Pure Pursuit) due to its ability to explicitly handle system constraints and anticipate future trajectory deviations. The tuning parameters adopted for the MPC were:

$$k_x = 50, \quad k_y = 50, \quad k_\psi = 10$$

$$W_{terminal} = 50, \quad W_{steering} = 10000$$

Initial tests were conducted using a kinematic model for the MPC. With this formulation, the vehicle successfully reached a maximum speed of 23 m/s (Fig. 11). However, at 25 m/s, it became infeasible to complete the lap while maintaining the lateral error within the 1 m constraint (Fig. 12).

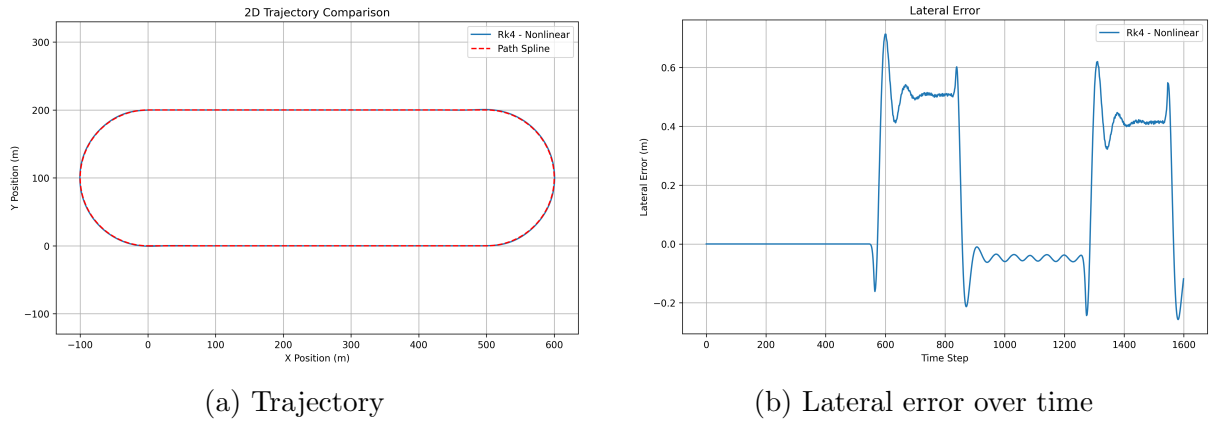


Figure 11: Kinematic MPC test at 23 m/s

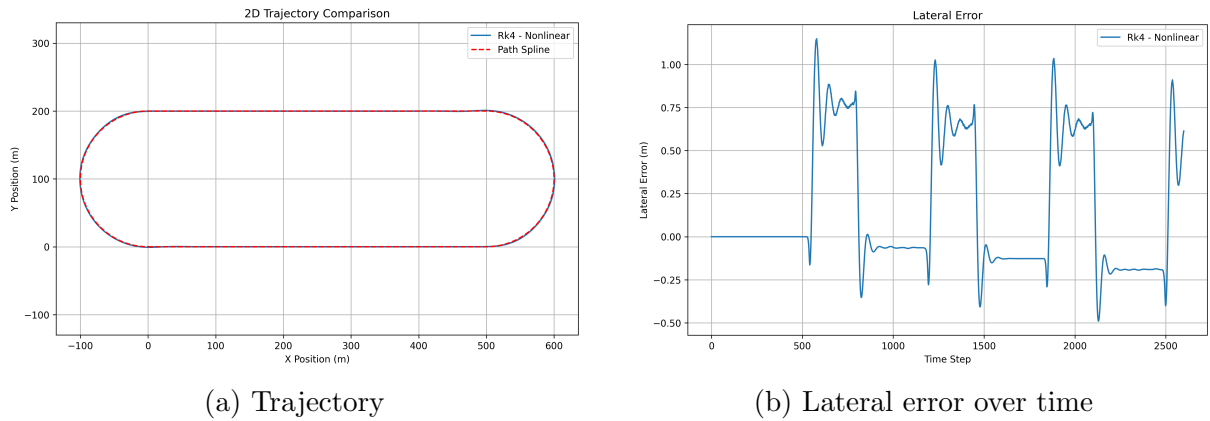


Figure 12: Kinematic MPC test at 25 m/s

The kinematic model was subsequently replaced by a linear Single-Track model, which is capable of better predicting the vehicle's behavior in high-speed scenarios by accounting for lateral tire forces and slip angles. This enhancement was expected to allow for higher velocities, a hypothesis confirmed by the results: the implementation enabled the vehicle to reach a maximum speed of 29.6 m/s while satisfying all constraints (Fig. 13). It is

likely that even higher speeds could have been achieved using a non-linear Single-Track model; however, this has not been implemented in this assignment.

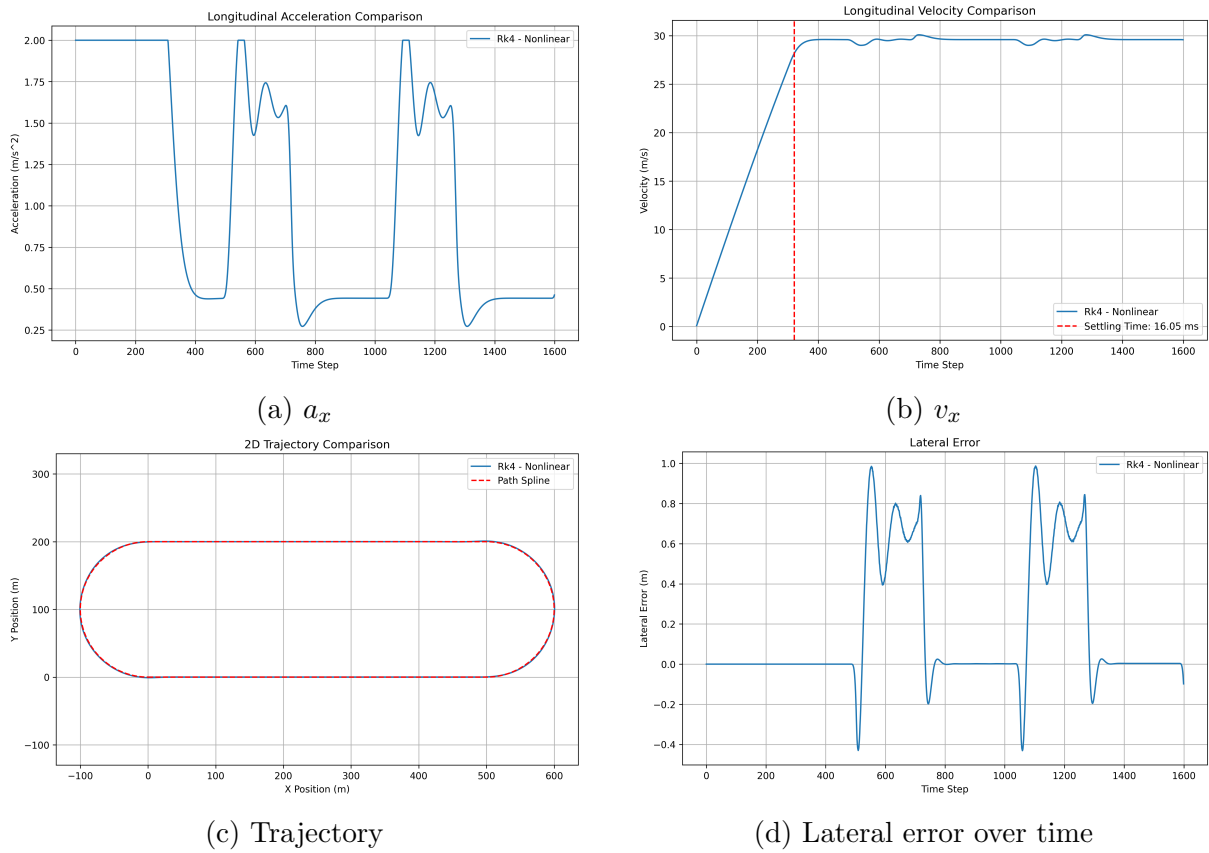
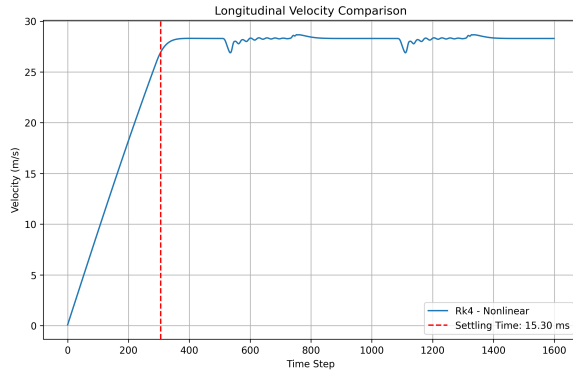


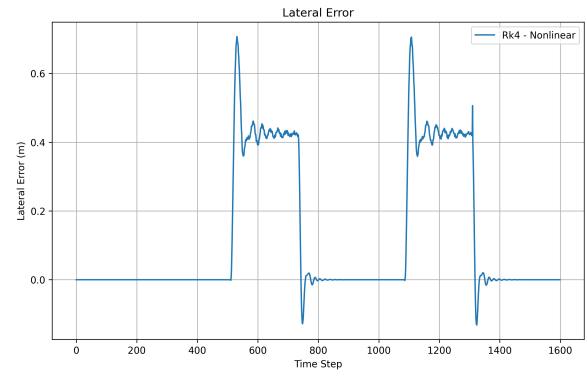
Figure 13: MPC linear Single-Track model results

3.3 Maximum velocities achieved

The three controllers achieved different maximum velocities while satisfying the assignment requirements. Specifically, the Pure Pursuit controller respected the constraints up to a speed of 28.3 m/s (Fig. 14), while the Stanley controller achieved a maximum speed of 26.4 m/s (Fig. 15). The linear MPC outperformed both, reaching a top speed of 29.6 m/s (Fig. 13b). This outcome is a direct consequence of the phenomena discussed in Section 3.1: the smoother steering action of the Pure Pursuit algorithm minimizes induced tire drag, allowing for a higher top speed and reduced longitudinal velocity error compared to the Stanley controller, whose high-frequency corrections dissipate more energy. Furthermore, the linear MPC controller outperforms both kinematic algorithms thanks to its path prediction capability. This results in smoother control inputs and a higher overall vehicle speed.

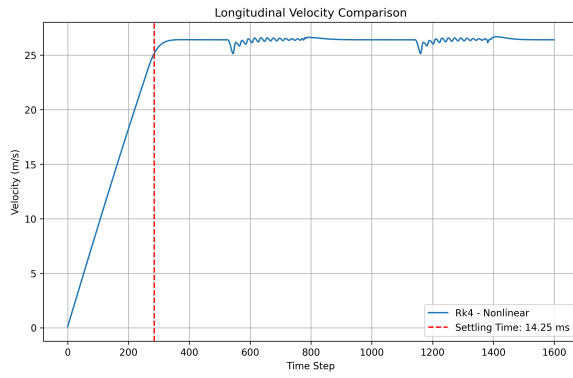


(a) Longitudinal Velocity

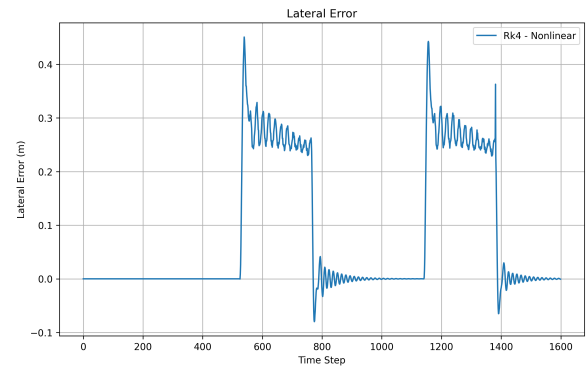


(b) Lateral error over time

Figure 14: Max velocity obtained by Pure Pursuit (28.3 m/s)



(a) Longitudinal Velocity



(b) Lateral error over time

Figure 15: Max velocity obtained by Stanley (26.4 m/s)